RAS: Retrieval-And-Structuring for Knowledge-Intensive LLM Generation

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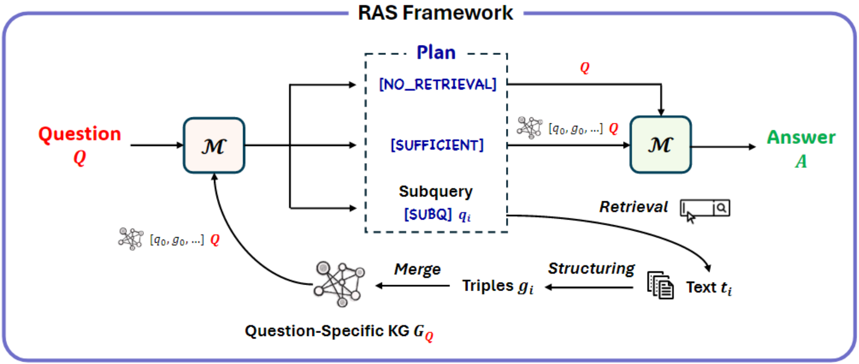
ICLR 2026

Outline

RAG and GraphRAG

- 1.In knowledge-intensive tasks, LLMs must organize evidence for multi-step reasoning. In open-domain and multi-hop QA, models need to connect multiple facts rather than rely on a single passage.
- 2.Traditional RAG feeds retrieved results as flat text chunks into models, without modeling entity relations, factual connections, or reasoning paths. This causes unstable reasoning and interference from irrelevant results.
- 3.GraphRAG uses knowledge graphs to enhance multi-hop reasoning, but suffers from high construction/maintenance costs and noise from global graphs during retrieval.
- 4.Our proposed RAS framework dynamically retrieves and constructs a local question-relevant knowledge graph to improve retrieval and reasoning performance.

RAS Framework



Knowledge-aware Planning

- 1.Initial Planning:

$$p_0 \leftarrow \mathcal{M}(\emptyset; \texttt{INST}_{\texttt{Plan}}; \emptyset; Q)$$

Output: [SUBQ], [NO_RETRIEVAL].

- 2.Iterative Planning:

$$p_{i+1} \leftarrow \mathcal{M}(\text{GNN}(G_i); \texttt{INST}_{\texttt{Plan}}; [q_0, g_0, ..., q_i, g_i]; Q)$$

Output: [SUBQ], [SUFFCIENT].

You are a planner to determine if the question can be answered with current information and output the appropriate label as well as the subquery if needed.
Output [NO_RETRIEVAL] if the question can be directly answered with the question itself without any retrieval.
Output [SUBQ] with an subquery for retrieval if still needs a subquery.
Output [SUFFICIENT] if the question can be answered with the provided information.

Figure 15: Instruction $\texttt{INST}_{\texttt{Plan}}$ for Planning.

Text Retrieval and Structuring

- 1.Text Retrieval: dense retriever (Contriever-MS MARCO, BM25).

$$t_i \leftarrow \text{Retrieval}(q_i, C, k)$$

- 2.Text-to-Triples Conversion: text-to-triples model f_{2t2} trained on WikiOFGraph dataset.

$$g_i \leftarrow f_{t2t}(t_i) = [(s_0, r_0, o_0), ..., (s_{|g_i|}, r_{|g_i|}, o_{|g_i|})]$$

Text:
"William Gerald Standridge (November 27, 1953 – April 12, 2014) was an American stock car racing driver. He was a competitor in the NASCAR Winston Cup Series and Busch Series."

Triples:
(S> William gerald standridge| P> Nationality| O> American),
(S> William gerald standridge| P> Occupation| O> Stock car racing driver),
(S> William gerald standridge| P> Competitor| O> Busch series),
(S> William gerald standridge| P> Competitor| O> Nascar winston cup series),
(S> William gerald standridge| P> Birth date| O> November 27, 1953),
(S> William gerald standridge| P> Death date| O> April 12, 2014)

Figure 13: Example of WikiOFGraph data

- 3.Knowledge Enrichment: Sentence-BERT embedding; merge evolving KG $G_Q = (V_Q, E_Q)$.

$$\text{emb}(v) \leftarrow \text{encode}(v), \forall v \in V_i; \qquad \text{emb}(e) \leftarrow \text{encode}(e), \forall e \in E_i$$

$$G_Q \leftarrow G_Q \cup g_i'$$

Knowledge-augmented Answering

- 1.$p_0 = [NO\_RETRIEVAL]$:

$$A \leftarrow \mathcal{M}(\emptyset; \mathtt{INST}_{\mathtt{Ans}}; \emptyset; Q)$$

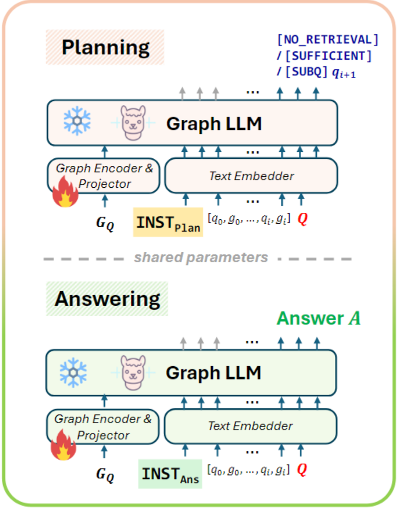
- 2.$p_i = [SUFFICIENT]$:

$$A \leftarrow \mathcal{M}(\mathrm{GNN}(G_Q); \mathtt{INST}_{\mathtt{Ans}}; [q_0, g_0, ..., q_i, g_i]; Q)$$

You are an answerer given a question and retrieved graph information.
Each [SUBQ] is a subquery we generated through reasoning for the question. The retrieved graph information follows each [SUBQ] is relevant graph information we retrieved to answer the subquery.
[NO_RETRIEVAL] means the question can be answered with the question itself without any retrieval.
The main question starts with "Question: ". Please answer the question, with subqueries and retrieved graph information if they are helpful.

Figure 16: Instruction $\mathtt{INST}_{\mathtt{Ans}}$ for Answering.

Structure-aware Multitask Learning



Training data

- HotpotQA-SUBQ: constructed based on HotpotQA; used to train planning and answering model *M*.
- 1.Filter out irrelevante "supporting docs".
- 2.Generating subqueries.
- 3.Sample labeling.
- 4.Incorporate the subset of Arc-Easy and ASQA (Self-RAG's training data).

Algorithm 1 HotpotQA-SUBQ Sample Labeling

Require: $\mathcal{D}$: Filtered HotpotQA dataset
Require: $\mathcal{M}$: Base LLM (LLaMA-2-7B)
Require: f_{t2t}: Text-to-triple conversion model
Ensure: $\mathcal{T}_{plan}$: Training data for Planning
Ensure: $\mathcal{T}_{ans}$: Training data for Answering

Initialize $\mathcal{T}_{plan}, \mathcal{T}_{ans} \leftarrow \{\}, \{\}$
for all $(Q, \{d_0, ..., d_n\}, A) \in \mathcal{D}$ **do**
  $\hat{A} \leftarrow \mathcal{M}(Q)$ ⟵ Direct answer attempt
  if $\hat{A} = A$ **then**
    $\mathcal{T}_{plan} \leftarrow \mathcal{T}_{plan} \cup \{(Q, \texttt{[NO\_RETRIEVAL]})\}$
    $\mathcal{T}_{ans} \leftarrow \mathcal{T}_{ans} \cup \{(Q, A)\}$
    continue
  end if
  $subq_0, ..., subq_n \leftarrow$ GenerateSubqueries$(Q, \{d_0, ..., d_n\})$
  $G_0, ..., G_n \leftarrow \emptyset$ ⟵ Initialize graph contexts
  for $i \leftarrow 0$ **to** n **do**
    $g_i \leftarrow f_{t2t}(d_i)$ ⟵ Convert text to triples
    if $i < n$ **then**
      $input \leftarrow$ FormatInput$(\{(subq_j, g_j)\}_{j=0}^{i}, Q)$
      $\mathcal{T}_{plan} \leftarrow \mathcal{T}_{plan} \cup \{(input, \texttt{[SUBQ]}\ subq_{i+1})\}$
    else
      $input \leftarrow$ FormatInput$(\{(subq_j, g_j)\}_{j=0}^{n}, Q)$
      $\mathcal{T}_{plan} \leftarrow \mathcal{T}_{plan} \cup \{(input, \texttt{[SUFFICIENT]})\}$
      $\mathcal{T}_{ans} \leftarrow \mathcal{T}_{ans} \cup \{(input, A)\}$
    end if
    $G_i \leftarrow G_{i-1} \cup g_i$ ⟵ Accumulate graph context
  end for
end for
return $\mathcal{T}_{plan}, \mathcal{T}_{ans}$

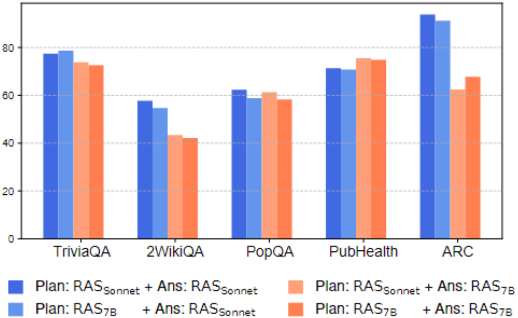
Result on benchmark

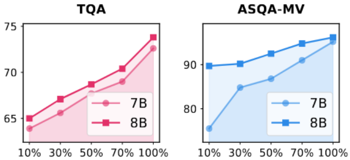
	Model/Method	Short-form TQA (acc)	Short-form 2WQA (F1)	Short-form PopQA (acc)	Closed-set Pub (acc)	Closed-set ARC (acc)	Long-form ASQA (rouge)	Long-form ASQA (mauve)	Long-form ELI5 (rouge)	Long-form ELI5 (mauve)
Closed-source	**w/o Retrieval**									
	ChatGPT	74.3	24.8	29.3	70.1	75.3	36.2	68.8	22.8	32.6
	Sonnet-3.5	78.4	40.0	30.2	83.7	88.5	37.0	39.1	21.8	26.5
	w/ Single Retrieval (#docs=5)									
	Sonnet-3.5$_{\#docs=1}$	69.1	41.9	51.5	49.1	88.6	n/a	n/a	n/a	n/a
	Sonnet-3.5$_{\#docs=5}$	72.5	53.7	57.3	53.9	87.1	38.8	61.6	20.2	32.3
	SuRe$_{\text{GPT-4o}}$(Kim et al., 2024b)	72.3	38.1	53.6	57.2	79.6	36.0	74.2	19.2	51.6
	SuRe$_{\text{Sonnet-3.5}}$	76.8	37.6	41.2	62.8	91.6	30.2	69.9	15.4	27.2
	w/ Self-Reflective Retrieval									
	ReAct$_{\text{Sonnet-3.5}}$ (Yao et al., 2023)	73.4	53.7	55.0	62.2	89.2	38.8	61.6	20.2	32.3
	IRCoT$_{\text{Sonnet-3.5}}$ (Trivedi et al., 2023)	74.7	54.9	53.2	59.4	92.0	38.8	61.6	20.2	32.3
	Retrieval-And-Structuring (ours)									
	RAS$_{\text{Sonnet-3.5}}$	77.6	57.7	62.3	71.3	93.9	39.1	70.5	23.3	37.7
Open-source	**w/o Retrieval**									
	Llama2$_{7B}$	30.5	18.9	14.7	34.2	21.8	15.3	19.0	18.3	32.4
	Llama2$_{13B}$	38.5	20.2	14.7	29.4	29.4	12.4	16.0	18.2	41.4
	Llama3$_{8B}$	56.1	21.2	26.7	33.2	42.2	17.6	25.0	18.2	39.7
	w/ Single Retrieval (#docs=5)									
	Llama2$_{7B}$	42.5	21.0	38.2	30.0	48.0	22.1	32.0	18.6	35.3
	Llama2$_{13B}$	47.0	31.2	45.7	30.2	26.0	20.5	24.7	18.6	42.3
	Llama3$_{8B}$	60.4	33.4	48.6	36.5	40.1	23.9	52.1	18.8	40.7
	SuRe$_{7B}$ (Kim et al., 2024b)	51.2	20.6	39.0	36.2	52.7	35.8	76.2	16.1	26.6
	w/ Self-Reflective Retrieval									
	Self-RAG$_{7B}$ (Asai et al., 2023)	66.4	25.1	54.9	72.4	67.3	35.7	74.3	17.9	35.6
	Self-RAG$_{13B}$	69.3	26.9	55.8	74.5	73.1	37.0	71.6	18.7	38.5
	RPG$_{7B}$ (Lyu et al., 2024b)	65.1	33.6	56.0	73.4	65.4	37.6	84.4	19.1	46.4
	ReAct$_{7B}$ (Yao et al., 2023)	64.0	25.0	42.7	52.4	59.0	22.1	32.0	18.6	35.3
	IRCoT$_{7B}$ (Trivedi et al., 2023)	61.5	27.6	44.3	59.6	61.6	22.1	32.0	18.6	35.3
	Retrieval-And-Structuring (ours)									
	RAS$_{7B}$	72.7	42.1	58.3	74.7	68.5	37.2	95.2	19.7	47.8
	RAS$_{8B}$	73.8	44.2	57.7	77.6	71.4	37.6	96.2	20.1	54.4

Ablation study

	TQA (acc)	2WQA (F1)	Pub (acc)	ASQA (rg)	ASQA (mv)
RAS_{7B}	**72.7**	**42.1**	**74.7**	**37.2**	**95.2**
Training Phase					
w/o GraphEncode	70.2	38.4	66.4	33.1	85.0
w/o LoRA	71.5	37.8	54.8	32.8	84.8
w/o Text-to-Triple	70.4	38.2	71.4	36.2	73.8
w/o Multi-Task	68.6	39.2	65.5	36.7	88.9
Inference Phase					
w/o Retrieval	56.9	27.4	69.0	31.3	70.6
w/o GraphEncode	68.8	38.7	67.3	36.5	93.6
w/o Planning	66.7	37.8	71.5	37.2	95.2

Component Efficiency





Triple Extractor f_{t2t}	Performance TQA	2WQA	PopQA	Efficiency (tokens/s)
Flan-T5-Large	70.3	40.7	56.7	166.0
<u>LLaMA-3.2-3B</u>	<u>72.7</u>	<u>42.1</u>	<u>58.3</u>	**4,885.3**
Claude-3.5-Sonnet	**73.8**	**44.5**	**60.1**	68.2